POINTED FORM OF THE QUASI-FINITE SPLITTING: AN ETALE NEIGHBOURHOOD WHOSE PULLBACK HAS ONE FINITE PIECE THROUGH THE GIVEN POINT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7543-7561 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 7563-7572, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pointed sharpening of the parent lemma, deliberately collected as its own aggregate per the lane's distinct-statement rule (the existence clause for v is the added content).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma etale-splits-off-quasi-finite-part (the unpointed global splitting, previously collected or in queue); Properties of Spaces, lemma points-cartesian (points of fibre products of spaces); existence of etale scheme charts of spaces.

Source reference. Original lemma 02LP: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7472–7482. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03H4: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7572–7572. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $x \in |X|$ with image $y \in |Y|$. Assume that*

(1) *f is locally of finite type,*
(2) *f is separated, and*
(3) *f is quasi-finite at x.*

Then there exists an étale morphism $(U, u) \to (Y, y)$ of pointed algebraic spaces and a decomposition

$$U \times_Y X = W \amalg V$$

into open and closed subspaces such that the morphism $V \to U$ is finite and there exists a point $v \in |V|$ which maps to x in $|X|$ and u in $|U|$.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Pick a scheme U, a point $u \in U$, and an étale morphism $U \to Y$ mapping u to y. There exists a point $x' \in |U \times_Y X|$ mapping to x in $|X|$ and u in $|U|$ (Properties of Spaces, Lemma 03H4). To finish, apply Lemma 02LP to the morphism $U \times_Y X \to U$ and the point x'. It applies because U is a scheme and hence u comes from the monomorphism $\mathrm{Spec}(\kappa(u)) \to U$. □